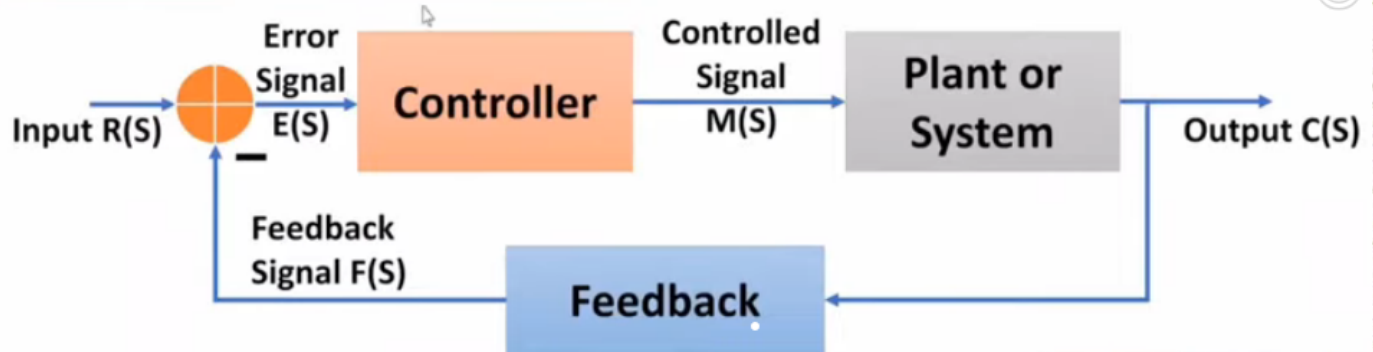


Controllers $\rightarrow$ In control system they are called decision makers.

They actually compare what you want (ref.) with what you actually get (O/p), and calculate error, then decide how much control action is required to apply so the system behaves properly.



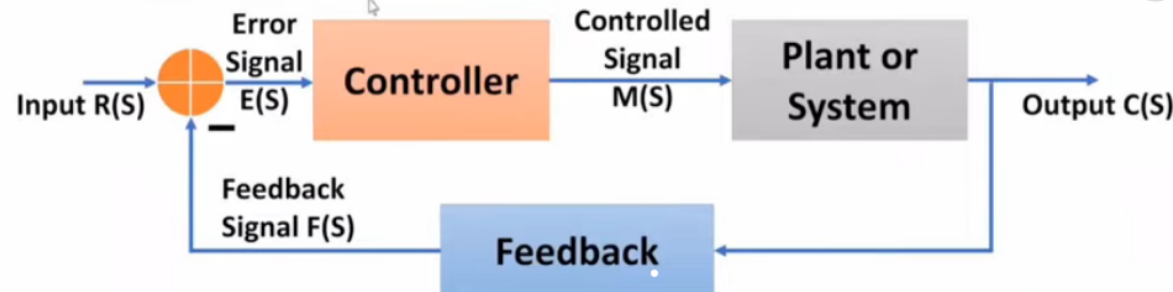
So a controller is a device that takes
 $\text{error} = \text{ref. value} - \text{actual value}.$

If the controller is a proportional controller,
then the gain of controller is constant k_p .

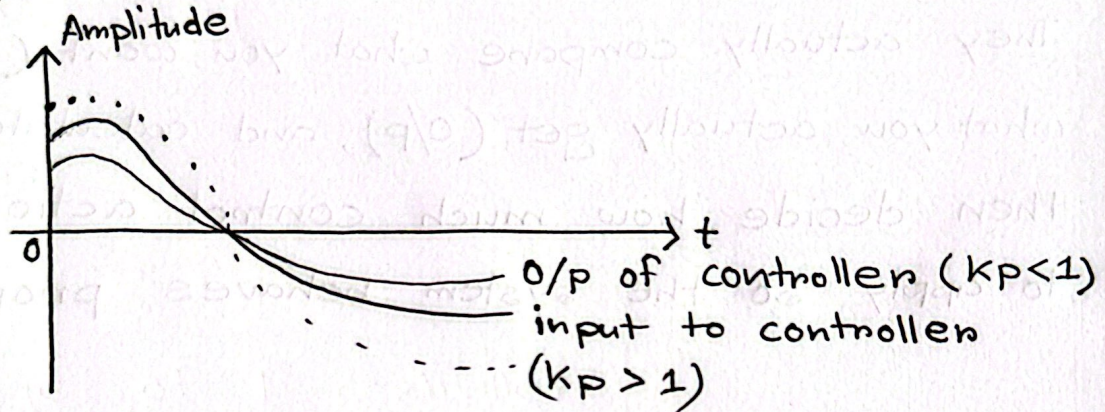
Let the system gain is $G(s)$ and feedback is
 $H(s)$, so output of P. controller is -

$$\begin{aligned} M(s) &= E(s) * k_p \\ &= (R(s) - F(s)) k_p \\ &= (R(s) - C(s) H(s)) k_p \end{aligned}$$

TF of the system is, $T(s) = \frac{C(s)}{R(s)} = \frac{k_p G(s)}{1 + k_p G(s) H(s)}$

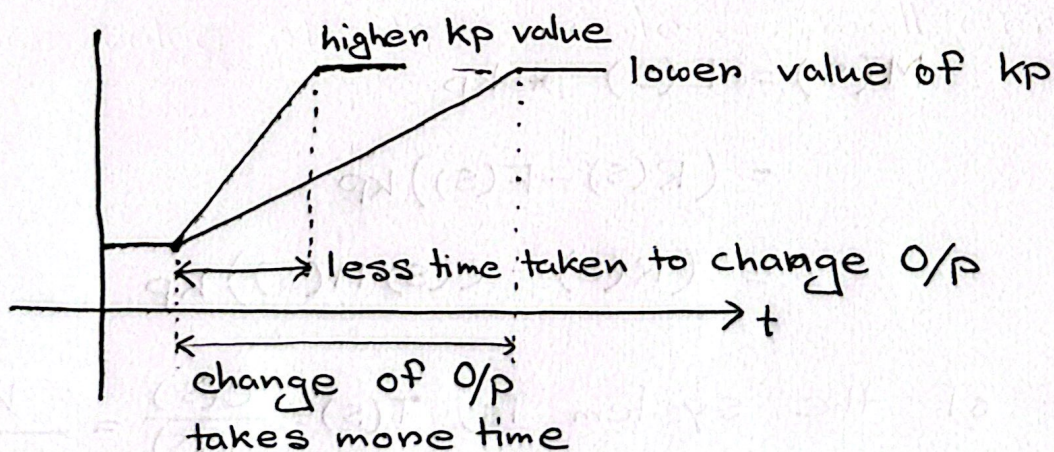


- * As gain of P. controller is constant, k_p , the shape of the output of the P. controller is same as the input to the controller. Only the output is multiplied by k_p .



Significance of P. Controller

- For higher value of k_p , the response is faster, that means it takes less time to change output.
- For lower value of k_p , the response is slower and more time is taken to get change in o/p.



Consider a 2nd order system

$$G(s) = \frac{1}{s^2 + as + b}$$

$$\begin{aligned} \text{T.F of this system, } T(s) &= \frac{k_p G(s)}{1 + k_p G(s)} \\ &= \frac{k_p}{s^2 + as + b + k_p} \quad \dots \textcircled{1} \end{aligned}$$

$$\text{Standard 2nd order eqn is } \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad \dots \textcircled{2}$$

Compare eqn $\textcircled{1}$ and $\textcircled{2}$

$$\omega_n = \sqrt{b + k_p} \quad \dots \textcircled{3}$$

$$2\xi\omega_n = a$$

$$\Rightarrow \xi = \frac{a}{2\sqrt{b + k_p}} \quad \dots \textcircled{4}$$

From eqn $\textcircled{3}$ we can conclude that, when k_p increases, natural frequency also increases. So, system becomes faster.

From eqn $\textcircled{4}$, when k_p rises, damping ratio, ξ decreases. So, system suffers from overshoot and oscillation.

- Q. Explain the effect of a P. controller on a 2nd order type-0 control system. In your answer, discuss how increasing the gain k_p affects system natural freq. response, damping ratio. Support your explanation with relevant mathematical expressions.

Advantages of P controller

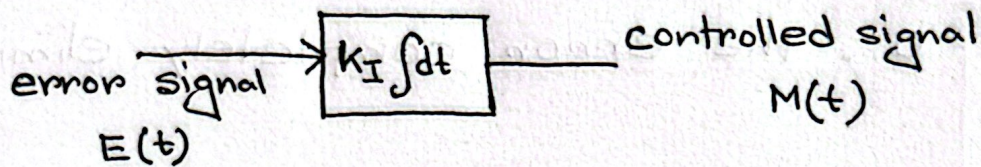
1. Simple to implement
2. Faster response

Disadvantages

1. This controller cannot eliminate SS error
2. Overshoot at higher k_p .
3. Low value of k_p creates small error.

Integral Controller: Control action $\propto$ accumulated error over time

It produces an output that is an integration of input signal.

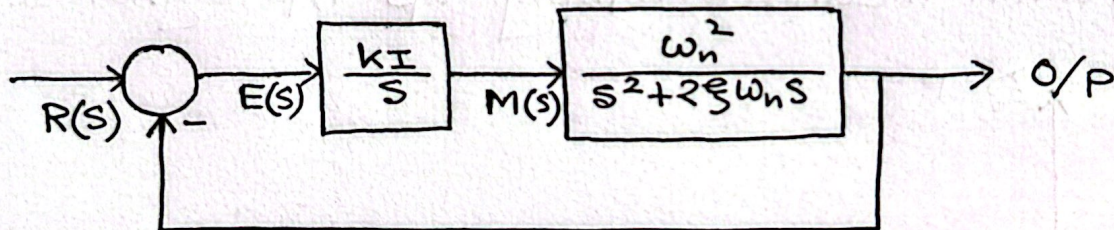


$$\therefore M(t) = k_I \int E(t) dt$$

Taking Laplace Transform - $M(s) = k_I \frac{E(s)}{s}$

$$\Rightarrow \frac{M(s)}{E(s)} = \frac{k_I}{s}$$

2nd order system:



$$T.F. = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} = \frac{k_I \omega_n^2}{s^3 + 2s^2 \xi \omega_n + k_I \omega_n^2} \quad \dots \textcircled{1}$$

General TF of 2nd order eqⁿ, $T.F. = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad \dots \textcircled{2}$

From eqⁿ ①, it is seen that as of I controller one pole is added in the system. This makes the decrease of system stability. As we know the more pole added in the system, the more system reaches to instability.

Before integral controller the system was type 0 type.

For step input, the $e_{ss} = \frac{1}{1+K_p}$. the e_{ss} was not 0.

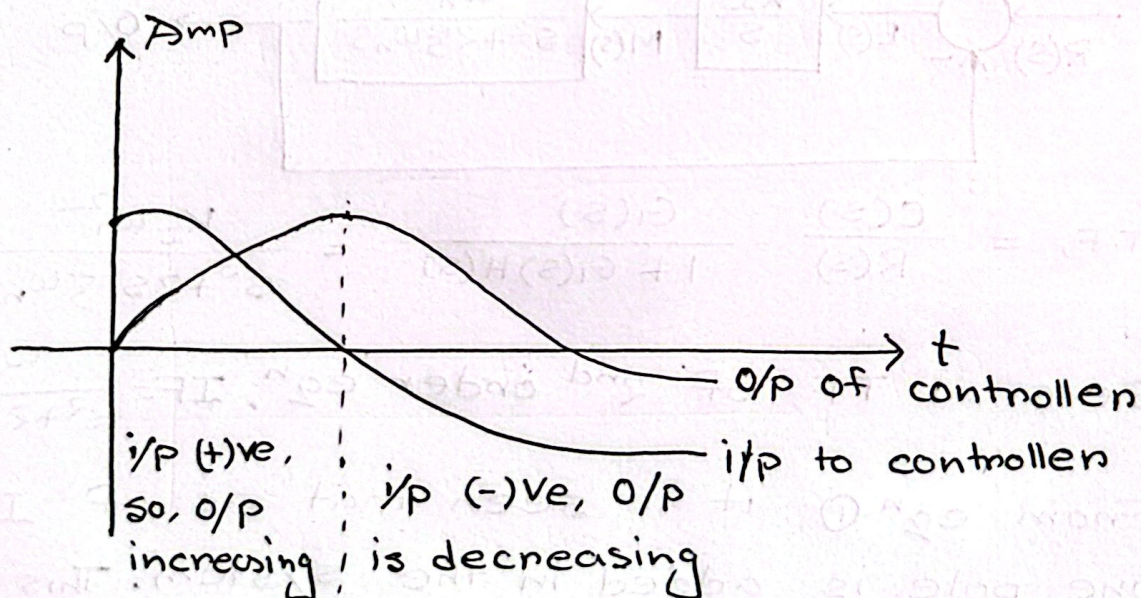
After adding I-controller the system became type 1 system.

So, $e_{ss} = 0$, the error completely eliminated.

Q. A 2nd order type 0 system is controlled using I-controller. Explain how the system order, steady state error and stability are affected.

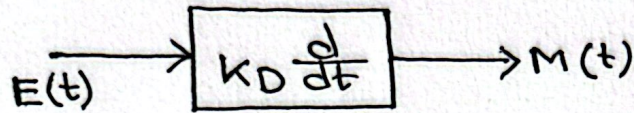
Understanding of I-controller:

- Integration means summing up i/p signal w.r.to time,



Derivative Controller: control action $\propto$ rate of change of error

So, derivative controller produces an O/p that is differentiation of input signal.

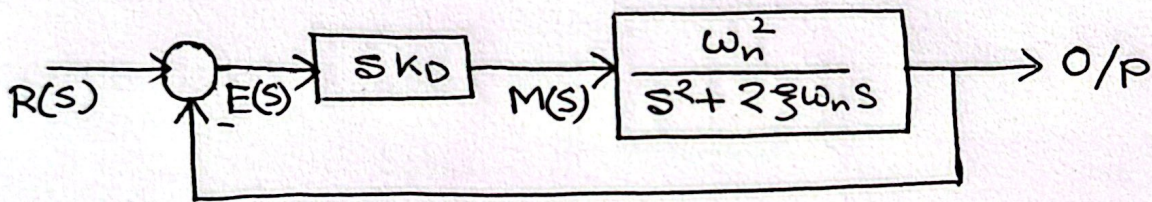


$$M(t) = k_D \frac{dE(t)}{dt}$$

Taking Laplace Transform: $M(s) = s k_D E(s)$

$$\Rightarrow \frac{M(s)}{E(s)} = s k_D$$

2nd Order system

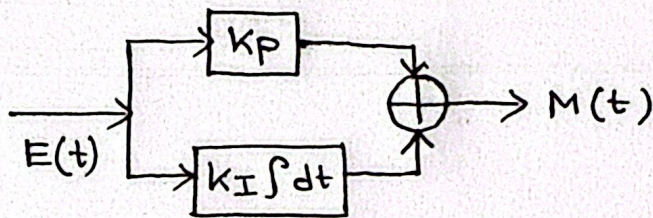


$$TF = \frac{s k_D G(s)}{1 + s k_D G(s)} = \frac{k_D \omega_n^2}{s + 2 \zeta \omega_n + k_D \omega_n^2}$$

Standard 2nd order system, $TF = \frac{\omega_n^2}{s^2 + 2 \zeta \omega_n s + \omega_n^2}$

A D-controller adds 0 at the origin, that improves system stability

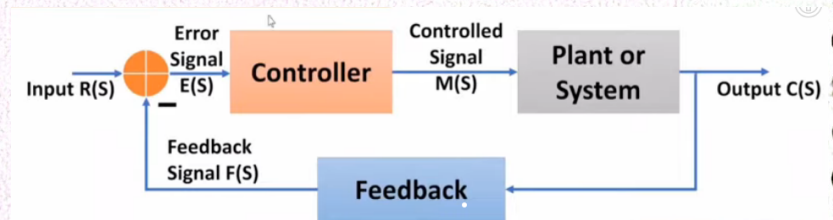
PI Controller:



$$M(t) = k_p E(t) + k_I \int E(t) dt$$

Laplace transform, $M(s) = k_p E(s) + k_I \frac{E(s)}{s}$

$$\therefore \frac{M(s)}{E(s)} = k_p + \frac{k_I}{s}$$



Significance:

- PI controller eliminates steady state error,
- Better stability offered compared to only I-controller
- Improved damping compared to P and PD

2nd order system:

$$G(s) = \frac{\omega_n^2}{s(s + 2\xi\omega_n)}$$

$$G_c(s) = k_p + \frac{k_I}{s}, H(s) = 1$$

$$TF = \frac{G(s)}{1 + G(s)H(s)}$$

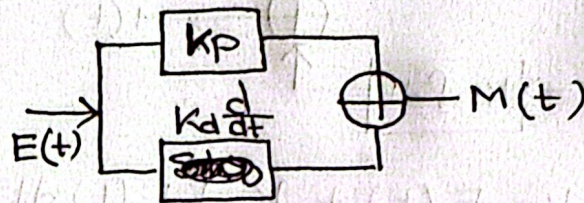
$$G_T(s) = \left(k_p + \frac{k_I}{s}\right) \frac{\omega_n^2}{s^2 + 2\xi\omega_n s}$$

$$= \frac{(k_I + s k_p) \omega_n^2}{s^2 + 2\xi\omega_n s}$$

$$= \frac{(k_I + s k_p) \omega_n^2}{s^2(s + 2\xi\omega_n) + (k_I + s k_p) \omega_n^2}$$

$$= \frac{(k_I + s k_p) \omega_n^2}{s^3 + 2\xi\omega_n s^2 + k_p \omega_n^2 s + k_I \omega_n^2}$$

PD Controller:



$$M(t) = E(t) \left[K_p + K_D \frac{d}{dt} \right]$$

$$= K_p E(t) + K_D \frac{dE(t)}{dt}$$

$$\therefore M(s) = K_p E(s) + s K_D E(s)$$

$$\Rightarrow \frac{M(s)}{E(s)} = K_p + s K_D$$

Significance:

- Error reduced due to P Controller.
- Reduced overshoot and oscillations.
- Improved stability due to D controller.

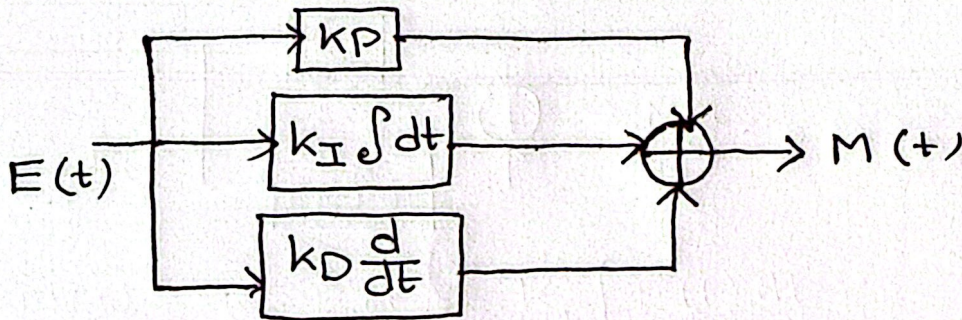
2nd order system:

$$G_c(s) = K_p + s K_D, \quad G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s}$$

$$TF = \frac{G_c(s)}{1 + G_c(s)G(s)} = \frac{(K_p + s K_D)\omega_n^2}{s^2 + 2\zeta\omega_n s + K_p\omega_n^2 + s K_D\omega_n^2}$$

as from TF it is seen that additional zero in the numerator, so system stability rises.

PID controller:



$$M(s) = \left(K_p + \frac{K_I}{s} + K_D s \right) E(s)$$

$$\Rightarrow TF = K_p + \frac{K_I}{s} + K_D s$$

Significance:

- No ss error due to I
- reduced overshoot and oscillation due to D.
- Faster response due to P and D.
- Improved stability ~~de~~ due to D

2nd order system:

$$\frac{G(s)}{1 + G(s)H(s)} = TF$$

$$G(s) = \left(K_p + \frac{K_I}{s} + K_D s \right) \left(\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s} \right)$$
$$= \frac{\omega_n^2 \left(K_p + \frac{K_I}{s} + K_D s \right)}{s^2 + 2\zeta\omega_n s}$$

$$\Rightarrow TF = \frac{\omega_n^2 \left(K_p + \frac{K_I}{s} + K_D s \right)}{s^3 + s^2(2\zeta\omega_n + K_D\omega_n^2) + s(\omega_n^2 K_p) + K_I\omega_n^2}$$

